Here petrol fuel air mixture is not sent in to the cylinder through intake manifold and inlet valve. In case of that only air is inducted to cylinder through the intake manifold, whereas petrol fuel was directly injected in to the cylinder. Because of this engine consumes less fuel and produces higher performance. But presently this type of injection system was not in practice, because of some uncontrollable parameters such as exhaust smoke and petrol fuel consumption.

b) Single point or throttle body injection:

In this system injectors are placed at each and every throat in the throttle body. This injector injects and sprays the petrol fuel before the throttle valve in the air. This developed spray of fuel was sent to the intake manifold and mixed with the air. By this way air fuel mixture is sent to the intake port through the inlet manifold. Non-uniform rate of air fuel mixture, vaporization of petrol fuel in the intake manifold, pressure difference inside the manifold are some of the problems involved in this system.

c) Multi –point or port injection:

In this system a separate injector was mounted in to the each and every cylinders intake manifold line. This injector injects and sprays petrol fuel in to the air passage in the intake manifold. This process denotes multi point fuel injection system.

d) Timed and continuous fuel injection:

Injection was depends on timed control it denotes this timed fuel injection. Fuel injection occurs from the injector by fixed timing mode in this system. In continuous injection system fuel injection occurs continuously in to the manifold. The cost and handling in continuous injection system is low and easy compared to timed injection system. In this system more amount of fuel injects in to the intake manifold with low pressure. The injected fuel mixed with the air in the manifold and mixed to form flammable mixture before entering in to the intake port. Now this mixture is ready to inject in to each cylinder. These mixtures get in to the engine by the opening of each and every inlet valve respectively. Inside the cylinder these mixture get mixed more fine for attaining its ignition point. In timed injection system fuel get injected in to the manifold at particular duration given from the timed control unit. In this system by controlling the fuel supply, the consumption of petrol fuel gets controlled. This system looks like diesel injection system used in the diesel engine.

9.7 COMPARISON BETWEEN MPFI AND CARBURETTOR

The working principle of MPFI and Carburettor are mostly same. However, there are several methods are used to identify the amount of air required when sending fuel in to the engine

Carburettor:

When engine is running under idle or slow speed the throttle valve is in closed position, hence the amount of air inducted in to the engine was measured using the pressure difference occurred in the manifold surroundings. During normal running condition the Vacuum created inside the venturi is used to measure the amount of air supplied in to the engine

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